- The mean of y_t exists if $\phi_1 \neq 1$.
- The mean of y_t is zero iff $\phi_0 = 0$.
- The constant term ϕ_0 relates to the mean of y_t by $\phi_0 = (1 - \phi_1)\,\mu$.

- Consider the variance of y_t:

$$Var\,(y_t) = \phi_1^2 Var\,(y_{t-1}) + Var\,(\varepsilon_t) + 2Cov\,(y_{t-1}, \varepsilon_t)\,. \tag{11}$$

- Using $\phi_0 = (1 - \phi_1)\,\mu$, model (10) can be rewritten as

$$y_t - \mu = \phi_1\,(y_{t-1} - \mu) + \varepsilon_t.$$

Let $\tilde{y}_t = y_t - \mu$. Then $\tilde{y}_t = \phi_1 \tilde{y}_{t-1} + \varepsilon_t$. By recursive substitutions,

$$\tilde{y}_t = \phi_1^k \tilde{y}_{t-k} + \phi_1^{k-1} \varepsilon_{t-k+1} + \ldots + \phi_1^2 \varepsilon_{t-2} + \phi_1 \varepsilon_{t-1} + \varepsilon_t.$$

If $|\phi_1| < 1$, then $\lim_{k\to\infty} \phi_1^k \tilde{y}_{t-k} = 0$, and

$$\tilde{y}_t = \varepsilon_t + \phi_1 \varepsilon_{t-1} + \phi_1^2 \varepsilon_{t-2} + \ldots = \sum_{i=0}^{\infty} \phi_1^i \varepsilon_{t-i}. \tag{12}$$

Therefore, $Cov\,(y_{t-1}, \varepsilon_t) = Cov\,(\mu + \sum_{i=0}^{\infty} \phi_1^i \varepsilon_{t-1-i}, \varepsilon_t) = 0$, and we obtain from (11) that

$$Var\,(y_t) = \frac{\sigma^2}{1 - \phi_1^2}, \quad \text{if } \phi_1^2 < 1.$$

- In order for $Var\,(y_t)$ to be bounded and non-negative, we must have $\phi_1^2 < 1$, or $|\phi_1| < 1$.
- If $|\phi_1| < 1$, it can be seen from (12) that $E\,(y_t) = \mu$ and $Var\,(y_t) = \sigma^2 \sum_{i=1}^{\infty} \phi_1^{2i} = \sigma^2 / (1 - \phi_1^2)$, that is, the mean and variance of y_t are finite and time invariant.
- If $|\phi_1| < 1$, then $AR\,(1)$ can be expressed as an $MA\,(\infty)$.

- Consider now the ACF of y_t. Without loss of generality, we abstract the mean from y_t

(set $\phi_0 = 0$). One approach to derive the ACF is by using the $MA(\infty)$ representation of an $AR(1)$. Recall that for an $MA(\infty)$ process, $y_t = \sum_{i=0}^{\infty} \theta_i \varepsilon_{t-i}$, its autocovariance is $\gamma_h = \sigma^2 \sum_{i=0}^{\infty} \theta_i \theta_{i+|h|}$. It follows that

$$\begin{aligned}
\gamma_0 &= \sigma^2 \left(\sum_{i=0}^{\infty} \phi_1^{2i} \right) = \frac{\sigma^2}{1-\phi_1^2}, \quad \rho_0 = 1, \\
\gamma_1 &= \sigma^2 \left(\sum_{i=0}^{\infty} \phi_1^i \phi_1^{i+1} \right) = \sigma^2 \phi_1 \sum_{i=0}^{\infty} \phi_1^{2i} = \frac{\phi_1 \sigma^2}{1-\phi_1^2}, \quad \rho_1 = \phi_1, \\
&\vdots \\
\gamma_h &= \frac{\phi_1^h \sigma^2}{1-\phi_1^2}, \quad \rho_h = \phi_1^h.
\end{aligned}$$

The ACF of a weakly stationary $AR(1)$ process decays exponentially at the rate of ϕ_1 (from the starting value $\rho_0 = 1$).

- An alternative method to find the ACF is simpler and turns out to be especially helpful when analyzing higher-order autoregressive processes later.

$$\begin{aligned}
\gamma_1 &= E(y_{t-1} y_t) = E[y_{t-1}(\phi_1 y_{t-1} + \varepsilon_t)] = \phi_1 E\left(y_{t-1}^2\right) + E(y_{t-1}\varepsilon_t) = \phi_1 \gamma_0, \\
&\Rightarrow \rho_1 = \phi_1, \\
\gamma_2 &= E(y_{t-2} y_t) = E[y_{t-2}(\phi_1 y_{t-1} + \varepsilon_t)] = \phi_1 E(y_{t-2} y_{t-1}) + E(y_{t-2}\varepsilon_t) = \phi_1 \gamma_1 = \phi_1^2 \gamma_0, \\
&\Rightarrow \rho_2 = \phi_1^2, \\
&\vdots \\
\gamma_h &= E(y_{t-h} y_t) = E[y_{t-h}(\phi_1 y_{t-1} + \varepsilon_t)] = \phi_1 \gamma_{h-1} = \phi_1^h \gamma_0, \\
&\Rightarrow \rho_h = \phi_1^h,
\end{aligned}$$

where we have used that $E(y_{t-h}\varepsilon_t) = 0$, for $h > 0$.

- Using the lag operator, the $AR(1)$ model, $y_t = \phi y_{t-1} + \varepsilon_t$, can be written as

$$y_t = \phi L y_t + \varepsilon_t, \quad \text{or} \quad (1 - \phi L) y_t = \varepsilon_t.$$

If $|\phi| < 1$, we can "invert" the $AR(1)$ and write

$$y_t = (1 - \phi L)^{-1} \varepsilon_t = \left(\frac{1}{1 - \phi L} \right) \varepsilon_t,$$

where

$$(1 - \phi L)^{-1} = 1 + \phi L + \phi^2 L^2 + \phi^3 L^3 + \ldots.$$

Note that

$$(1 - \phi L)^{-1} (1 - \phi L) = 1,$$

where "1" denotes the identity operator, namely, $1 y_t = y_t$. Then

$$y_t = (1 - \phi L)^{-1} \varepsilon_t = \left(1 + \phi L + \phi^2 L^2 + \phi^3 L^3 \ldots \right) \varepsilon_t = \sum_{i=0}^{\infty} \phi^i \varepsilon_{t-i}.$$

5.2 AR(2) Model

- Consider the $AR(2)$ model:

$$y_t = \phi_0 + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \varepsilon_t, \tag{13}$$

and suppose $\{y_t\}$ is weakly stationary.

Exercise: Show that

$$E(y_t) = \mu = \frac{\phi_0}{1 - \phi_1 - \phi_2}, \quad \text{if } \phi_1 + \phi_2 \neq 1,$$
$$\rho_0 = 1, \quad \rho_1 = \frac{\phi_1}{1 - \phi_2},$$
$$\rho_h = \phi_1 \rho_{h-1} + \phi_2 \rho_{h-2}, \ h \geq 2.$$

- The autocovariances and autocorrelations follow the same *linear second-order difference equation* as does $\{y_t\}$, with the difference equation for γ_h and ρ_h indexed by the lag h.[2] Using the lag operator notation,

$$\left(1 - \phi_1 L - \phi_2 L^2\right) y_t = \phi_0 + \varepsilon_t,$$
$$\left(1 - \phi_1 L - \phi_2 L^2\right) \rho_h = 0.$$

- Consider the second-order difference equation, $y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + u_t$. The *general solution* of y_t has the form

$$y_t = c_1 \lambda_1^t + c_2 \lambda_2^t,$$

where c_1 and c_2 are constants determined by given initial values, and λ_1 and λ_2 are roots (assumed to be distinct) of the *characteristic equation*

$$\lambda^2 - \phi_1 \lambda - \phi_2 = 0. \tag{14}$$

If $\lambda_1 = \lambda_2$, the solution takes more complex forms but the main conclusions are unaffected. For $\{y_t\}$ to be stationary, it is necessary and sufficient that the solutions of (14) *lie strictly inside the unit circle.*

- The solutions of (14) are

$$\lambda_{1,2} = \frac{\phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2}}{2}.$$

  - If $\phi_1^2 + 4\phi_2 > 0$, λ_1 and λ_2 are real numbers. For $\{y_t\}$ to be stationary, we must have $|\lambda_1| < 1$ and $|\lambda_2| < 1$. The ACF of y_t is a mixture of two exponential decays.
  - If $\phi_1^2 + 4\phi_2 < 0$, λ_1 and λ_2 are complex conjugates:[3]

$$\lambda_{1,2} = \frac{\phi_1 \pm i\sqrt{-(\phi_1^2 + 4\phi_2)}}{2},$$

[2] See Chapter 1 of Hamilton (1994) for a review of difference equations. You may also find this note on second-order difference equations helpful.

[3] For a review of trigonometry and complex numbers, see Hamilton (1994), Mathematical Review Appendix A.1 and A.2.

or equivalently

$$\lambda_1 = a + bi, \quad \lambda_2 = a - bi,$$

where $a = \phi_1/2$, $b = (1/2)\sqrt{-\phi_1^2 - 4\phi_2}$, and $i = \sqrt{-1}$. We can rewrite λ_1 and λ_2 in polar coordinate form:

$$\lambda_1 = R\left[cos\left(\theta\right) + i \cdot sin\left(\theta\right)\right],$$
$$\lambda_2 = R\left[cos\left(\theta\right) - i \cdot sin\left(\theta\right)\right],$$

where R and θ are defined by

$$R = \sqrt{a^2 + b^2} = \sqrt{\left(\frac{\phi_1}{2}\right)^2 - \frac{\phi_1^2 + 4\phi_2}{4}} = \sqrt{-\phi_2}.$$

which is the modulus of the complex number, and $cos\left(\theta\right) = a/R$, and $sin\left(\theta\right) = b/R$. Then

$$\lambda_1^t = R^t\left[cos\left(\theta t\right) + i \cdot sin\left(\theta t\right)\right],$$
$$\lambda_2^t = R^t\left[cos\left(\theta t\right) - i \cdot sin\left(\theta t\right)\right].$$

For $\{y_t\}$ to be stationary, we must have $R < 1$, or $\phi_2 > -1$. The ACF of y_t is a mixture of two damped sine and cosine waves.[4]

– Complex-valued characteristic roots are important for many economic and business applications since they give rise to business cycles. The average length of the stochastic cycles is given by

$$k = \frac{2\pi}{\theta} = \frac{2\pi}{cos^{-1}\left(a/R\right)} = \frac{2\pi}{cos^{-1}\left(\phi_1/2\sqrt{-\phi_2}\right)},$$

where cos^{-1} (or $arccos$) denotes the inverse of the cosine function.

[4] If λ_1 and λ_2 are complex conjugates, then c_1 and c_2 are complex conjugates, and hence $c_1\lambda_1^t + c_2\lambda_2^t$ is strictly real.

- *Exercise*: Determine whether the following $AR(2)$ processes are stationary.
  - (i) $y_t = 0.6y_{t-1} + 0.2y_{t-2} + \varepsilon_t$
  - (ii) $y_t = 0.5y_{t-1} - 0.8y_{t-2} + \varepsilon_t$

- Alternatively, letting $z = \lambda^{-1}$, stationarity requires that all roots of

$$1 - \phi_1 z - \phi_2 z^2 = 0$$

  lie *outside* the unit circle.

- Under the stationarity condition, we can "invert" $\left(1 - \phi_1 L - \phi_2 L^2\right)$ such that

$$\psi(L) = \left(1 - \phi_1 L - \phi_2 L^2\right)^{-1} = \psi_0 + \psi_1 L + \psi_2 L^2 + \psi_3 L^3 + \ldots,$$

  and

$$y_t = \psi(L)\phi_0 + \psi(L)\varepsilon_t, \tag{15}$$

  which is the $MA(\infty)$ representation of the $AR(2)$ process.
  - Factoring the lag polynomial

$$1 - \phi_1 L - \phi_2 L^2 = (1 - \lambda_1 L)(1 - \lambda_2 L),$$

where $\lambda_1\lambda_2 = -\phi_2$ and $\lambda_1 + \lambda_2 = \phi_1$; or $\lambda_{1,2} = \frac{\phi_1 \pm \sqrt{\phi_1^2+4\phi_2}}{2}$. Then

$$\begin{aligned}
\psi(L) &= (1-\lambda_1 L)^{-1}(1-\lambda_2 L)^{-1} \\
&= \left(\sum_{i=0}^{\infty} \lambda_1^i L^i\right)\left(\sum_{j=0}^{\infty} \lambda_2^j L^j\right) \\
&= 1 + (\lambda_1+\lambda_2)L + \left(\lambda_1^2 + \lambda_1\lambda_2 + \lambda_2^2\right)L^2 + \ldots \\
&= \sum_{i=0}^{\infty}\left(\sum_{j=0}^{i} \lambda_1^j \lambda_2^{i-j}\right)L^i.
\end{aligned}$$

  - It can be verified that $\psi(L)\phi_0 = \phi_0/(1-\phi_1-\phi_2)$ and $\sum_{i=0}^{\infty}|\psi_i| < \infty$. That is, (15) is an absolutely summable $MA(\infty)$ process.

- Alternative method to derive the $MA(\infty)$ representation of the $AR(2)$ process—*partial fraction decomposition (or method of undetermined coefficients):*

$$\frac{1}{(1-\lambda_1 L)(1-\lambda_2 L)} = \frac{a}{1-\lambda_1 L} + \frac{b}{1-\lambda_2 L},$$

which gives the polynomial identity

$$1 = a(1-\lambda_2 L) + b(1-\lambda_1 L),$$

or equivalently

$$a + b = 1,$$

$$\lambda_2 a + \lambda_1 b = 0.$$

Therefore,

$$a = \frac{\lambda_1}{\lambda_1 - \lambda_2},\ b = \frac{\lambda_2}{\lambda_2 - \lambda_1},$$

and it follows that

$$\begin{aligned} y_t &= \left(1-\phi_1 L-\phi_2 L^2\right)^{-1}\varepsilon_t = \frac{\lambda_1}{\lambda_1-\lambda_2}\left(\frac{1}{1-\lambda_1 L}\right)\varepsilon_t + \frac{\lambda_2}{\lambda_2-\lambda_1}\left(\frac{1}{1-\lambda_2 L}\right)\varepsilon_t \\ &= \frac{\lambda_1}{\lambda_1-\lambda_2}\sum_{i=0}^{\infty}\lambda_1^i\varepsilon_{t-i} + \frac{\lambda_2}{\lambda_2-\lambda_1}\sum_{i=0}^{\infty}\lambda_2^i\varepsilon_{t-i} \\ &= \sum_{i=0}^{\infty}\left(\frac{\lambda_1}{\lambda_1-\lambda_2}\lambda_1^i + \frac{\lambda_2}{\lambda_2-\lambda_1}\lambda_2^i\right)\varepsilon_{t-i}. \end{aligned}$$

5.3 AR(p) Model

- Consider the $AR(p)$ model:

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \ldots + \phi_p y_{t-p} + \varepsilon_t,$$

and suppose that $\{y_t\}$ is weakly stationary. Premultiplying both sides by $y_t, y_{t-1}, \ldots, y_{t-p}$ and taking expectations lead to

$$\begin{aligned} \gamma_0 &= \phi_1\gamma_1 + \phi_2\gamma_2 + \ldots + \phi_p\gamma_p + \sigma^2, \\ \gamma_1 &= \phi_1\gamma_0 + \phi_2\gamma_1 + \ldots + \phi_p\gamma_{p-1}, \\ &\ldots \\ \gamma_p &= \phi_1\gamma_{p-1} + \phi_2\gamma_{p-2} + \ldots + \phi_p\gamma_0, \end{aligned}$$

where we have used $E\left(y_{t-h}\varepsilon_t\right) = \sigma^2$, for $h = 0$; and $E\left(y_{t-h}\varepsilon_t\right) = 0$, for $h > 0$. The above system of equations is known as the *Yule-Walker equations*.

- Two usages of the Yule-Walker equations
  - Solving for γ from ϕ (recursively)

$$\begin{pmatrix} 1 & -\phi_1 & \ldots & -\phi_p \\ -\phi_1 & 1 & \ldots & -\phi_{p-1} \\ \vdots & \vdots & \ldots & \vdots \\ -\phi_p & -\phi_{p-1} & \ldots & 1 \end{pmatrix}\begin{pmatrix} \gamma_0 \\ \gamma_1 \\ \vdots \\ \gamma_p \end{pmatrix} = \begin{pmatrix} \sigma^2 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

– Estimating ϕ from γ

$$\begin{pmatrix} \gamma_0 & \gamma_1 & \cdots & \gamma_{p-1} \\ \gamma_1 & \gamma_0 & \cdots & \gamma_{p-2} \\ \vdots & \vdots & \ldots & \vdots \\ \gamma_{p-1} & \gamma_{p-2} & \cdots & \gamma_0 \end{pmatrix} \begin{pmatrix} \phi_1 \\ \phi_2 \\ \vdots \\ \phi_p \end{pmatrix} = \begin{pmatrix} \gamma_1 \\ \gamma_2 \\ \vdots \\ \gamma_p \end{pmatrix}$$

- Dividing both sides of the Yule-Walker equations by γ_0 yields the systems of equations for ACFs.
- The autocovariances and autocorrelations satisfy the same difference equation as y_t, and hence follow a mixture of damped exponentials and sine waves.
- $\{y_t\}$ is stationary if all the roots of the characteristic equation

$$\lambda^p - \phi_1 \lambda^{p-1} - \ldots - \phi_p = 0$$

lie *inside* the unit circle, or equivalently, all the roots of

$$1 - \phi_1 z - \ldots - \phi_p z^p = 0$$

lie *outside* the unit circle.

6 Autoregressive Moving Average Processes

6.1 ARMA(1,1) Model

- Consider the $ARMA\,(1,1)$ model (assuming for simplicity that $\phi_0 = 0$):

$$y_t = \phi y_{t-1} + \varepsilon_t + \theta \varepsilon_{t-1},$$

where $\varepsilon_t \sim WN\left(0, \sigma^2\right)$. Using the lag operator:

$$(1 - \phi L)\, y_t = (1 + \theta L)\, \varepsilon_t.$$

– The model is not identified if $\phi = -\theta$. If $|\phi| < 1$,

$$y_t = (1 - \phi L)^{-1} (1 + \theta L) \varepsilon_t = \left(\sum_{i=0}^{\infty} \phi^i L^i \right) (1 + \theta L) \varepsilon_t$$
$$= \varepsilon_t + (\phi + \theta) \varepsilon_{t-1} + \left(\phi^2 + \theta\phi \right) \varepsilon_{t-2} + \ldots + \left(\phi^i + \theta\phi^{i-1} \right) \varepsilon_{t-i} + \ldots.$$

– *Exercise*: Suppose that $\{y_t\}$ is weakly stationary. Show that

$$E(y_t) = 0, \quad Var(y_t) = \frac{(1 + 2\phi\theta + \theta^2)\sigma^2}{1 - \phi^2},$$
$$\rho_h = \frac{(\phi + \theta)(1 + \phi\theta)}{1 + 2\phi\theta + \theta^2} \phi^{h-1}, \ h \geq 1.$$

– The ACF of $ARMA(1,1)$ behaves similarly to that of $AR(1)$ except that the exponential decay begins at lag 2.

– Since the $MA(1)$ part of the process is stationary, y_t is stationary if the $AR(1)$ part of the process is stationary.

6.2 ARMA(p,q) Model

$$y_t = \sum_{i=1}^{p} \phi_i y_{t-i} + \sum_{i=0}^{q} \theta_i \varepsilon_{t-i}, \quad \theta_0 = 1, \tag{16}$$

or

$$\phi(L) y_t = \theta(L) \varepsilon_t,$$

where $\phi(L) = 1 - \phi_1 L - \ldots - \phi_p L^p$ is the AR polynomial, and $\theta(L) = 1 + \theta_1 L + \ldots + \theta_q L^q$ is the MA polynomial.